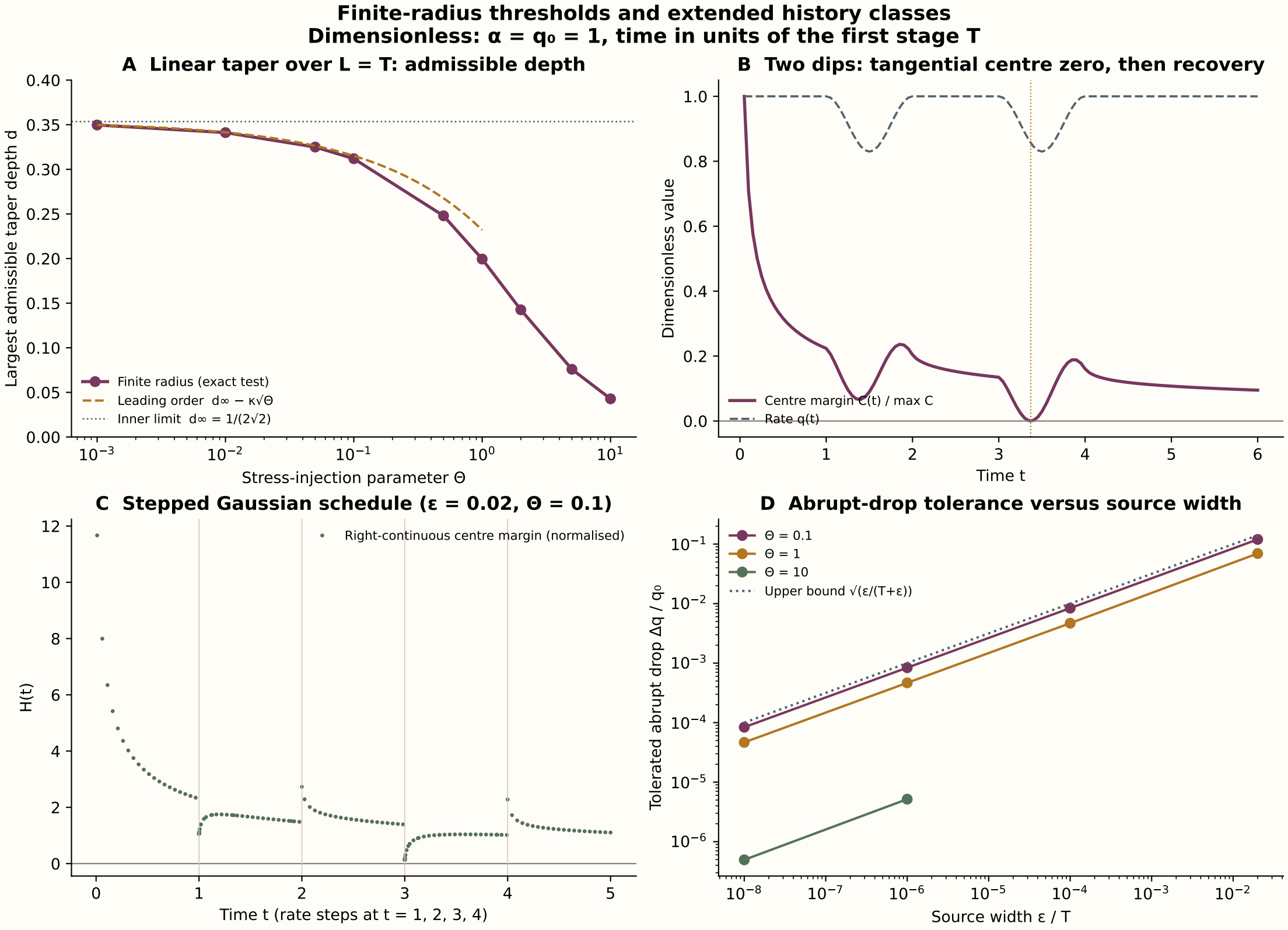
Finite-radius thresholds and extended history classes
Dimensionless: α = q₀ = 1, time in units of the first stage T
A Linear taper over L = T: admissible depth
Largest admissible taper depth d
Stress-injection parameter Θ
Finite radius (exact test)
Leading order d∞ − κ√Θ
Inner limit d∞ = 1/(2√2)
B Two dips: tangential centre zero, then recovery
Dimensionless value
Time t
Centre margin C(t) / max C
Rate q(t)
C Stepped Gaussian schedule (ε = 0.02, Θ = 0.1)
H(t)
Time t (rate steps at t = 1, 2, 3, 4)
Right-continuous centre margin (normalised)
D Abrupt-drop tolerance versus source width
Tolerated abrupt drop Δq / q₀
Source width ε / T
Θ = 0.1
Θ = 1
Θ = 10
Upper bound √(ε/(T+ε))